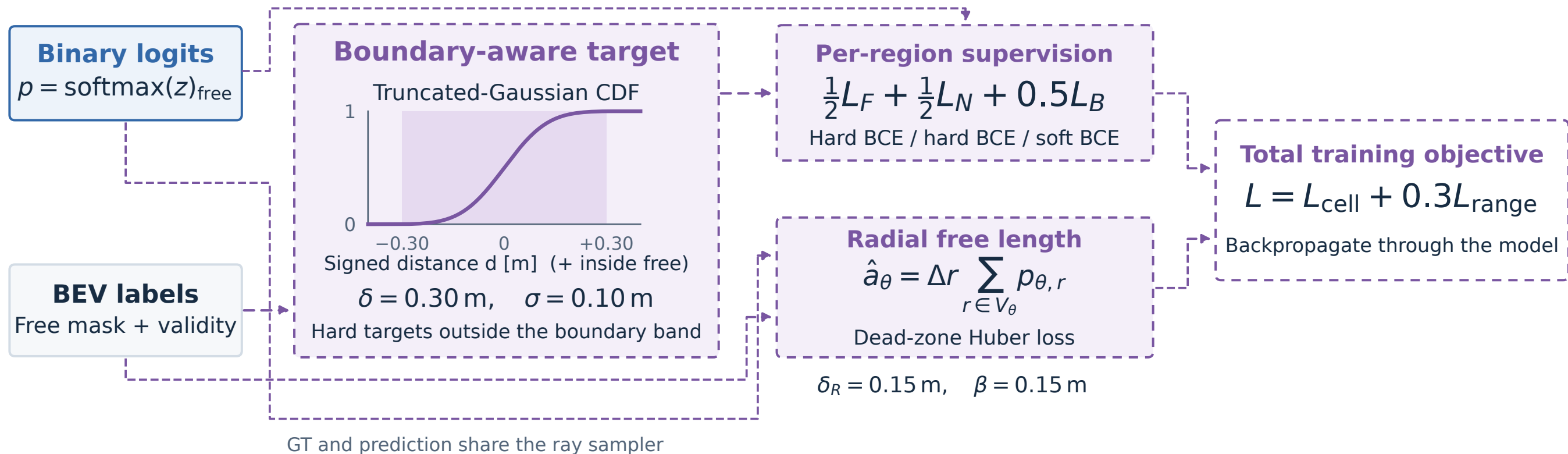


Training-only supervision

Same inference architecture for all loss ablations



Losses use valid cells; permanent blind cells receive hard not-free targets. Range loss uses eligible GT rays.